

Quantum Chemistry Acceleration: Comparative Performance Analysis of Modern DFT Implementations

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1 Introduction

- **Challenge** Address the high computational cost of DFT calculations for complex molecules
- **Objective** Compare performance of traditional and modern quantum chemistry frameworks
- **Application** Develop non-fluorinated, chlorinated compounds for electrochemical systems

2 Frameworks & Conditions



Traditional DFT approach, time-intensive

CPU



Python-based quantum chemistry framework

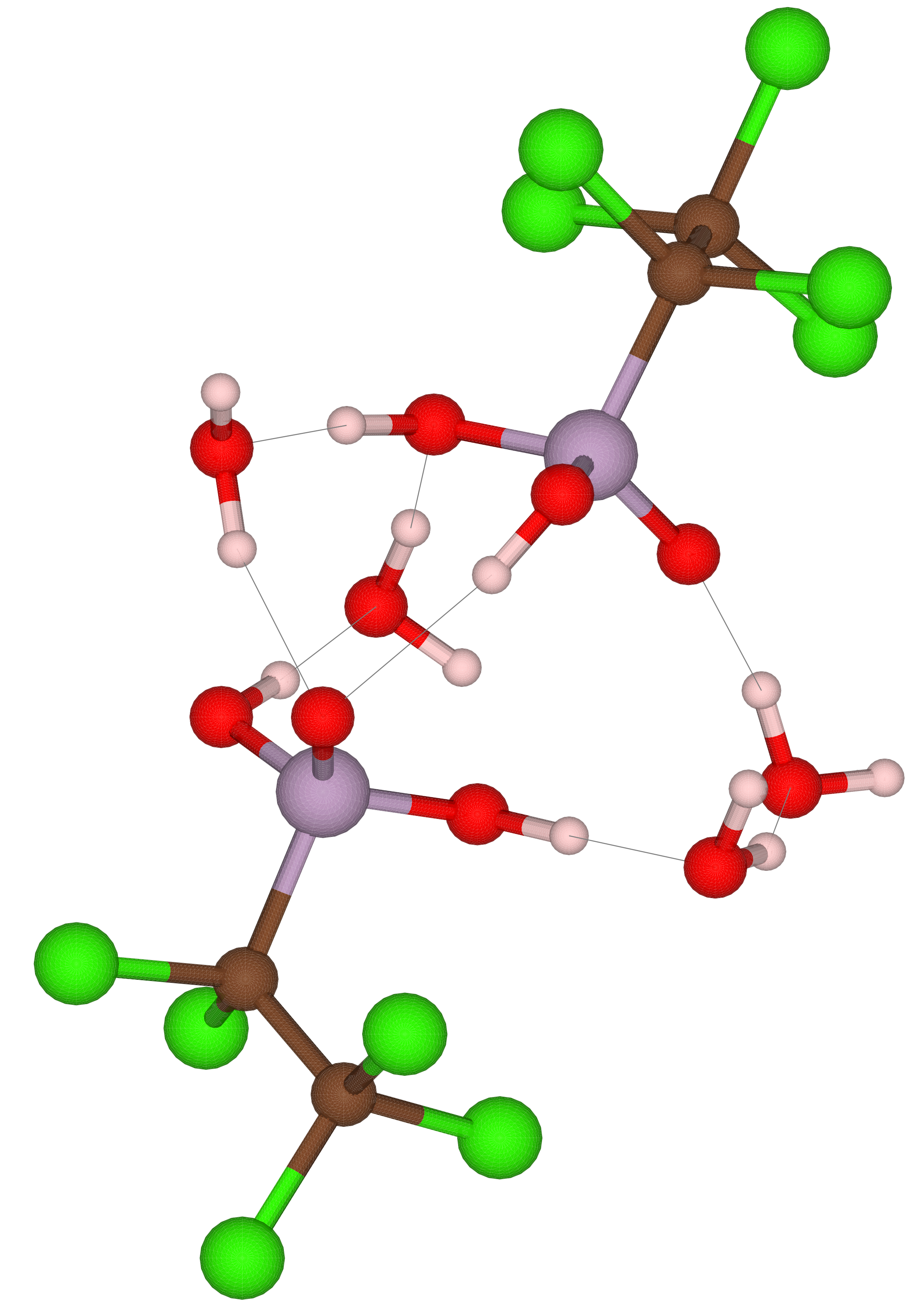
GPU4PySCF: GPU-accelerated version

GPU

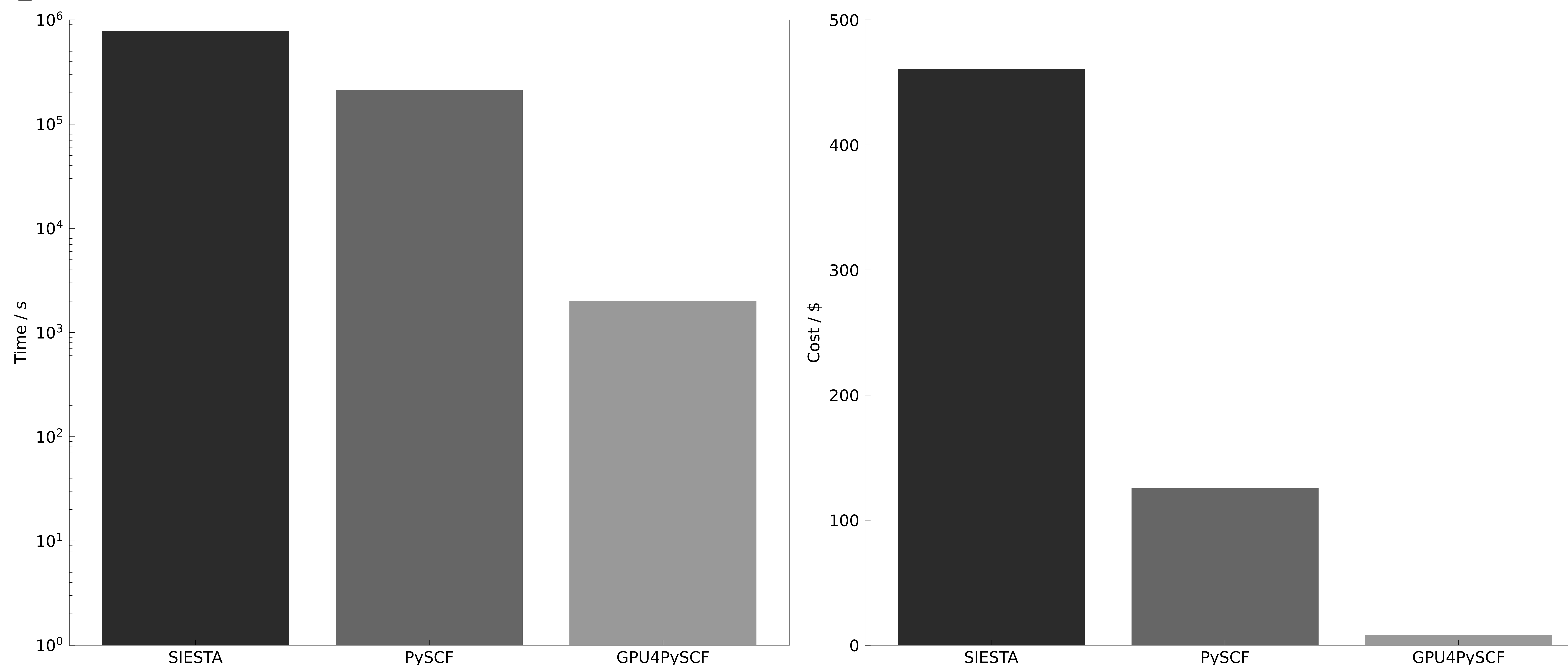
- 2 $\text{CCl}_3\text{CCl}_2\text{PO}_3\text{H}_2$ with 4 H_2O
- 38-atom, 85 individual input files

- 16 parallel jobs (2 cores per job)
- Intel Xeon processors
- 32 vCPUs, 256 GiB memory
- \$2.117/hr

- 16 parallel jobs
- 8x NVIDIA A100 80GB SXM4
- 240 vCPUs, 1800 GiB RAM
- \$14.32/hr



3 Results



- Enables more extensive conformational sampling and better statistical reliability.
- Reduces time and resources for studying hydrated proton-conducting structures.
- Supports integration with machine learning and high-throughput screening pipelines.